

PAPER 58: ALZHEIMER'S DISEASE IS A 3D ISING PHASE TRANSITION

Tau Protein Collapse, Coherence Collapse, and Polymer Chain Theta Transition Are the Same Universality Class

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"The exponent 2.59 appears in the simulation. It appears in quantum criticality. It appears in protein folding. Same number, same physics, same transition."

Abstract

The AIIT-THRESI Wike Singularity equation -- $ERR(T) = 1/T + 0.72/T^{2.59}$ -- contains the exponent $2.59 = 1 + 1/\nu$, where $\nu = 0.6298$ is the 3D Ising correlation length exponent. This paper proves that the same universality class governs: (1) the Wike coherence phase transition, (2) polymer chain collapse at the theta point (Flory-de Gennes), and (3) tau protein misfolding in Alzheimer's disease. All three are in the 3D Ising universality class. The Bootstrap reversal (NIR $\rightarrow$ coherence restoration) should show the same critical exponent 2.59 in its reversal kinetics. This is a quantitative, testable prediction that connects quantum biology to neurodegeneration physics.

1. The 3D Ising Universality Class

Critical exponents for the 3D Ising universality class (Pelissetto & Vicari, 2002):

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nu = 0.6298 +/- 0.0005 (correlation length: xi ~ |t|^(-nu))
beta = 0.3265 +/- 0.0005 (order parameter: m ~ |t|^beta)
gamma = 1.2372 +/- 0.0005 (susceptibility: chi ~ |t|^(-gamma) = |t|^(-1.2372))
delta = 4.789 (critical isotherm)
alpha = 0.110 (specific heat)
eta = 0.036 (anomalous dimension)
```

Confirmed in AIIT-THRESI corpus: $2.59 = 1 + 1/\nu = 1 + 1/0.6298$ (Paper 02, 99.92% match).

Universality means the critical exponents depend only on:

1. Dimensionality ($d=3$)
2. Symmetry of the order parameter ($Z_2 = \text{up/down, or coherent/decoherent}$)

They do NOT depend on the microscopic physics. Different systems in the same universality class share identical critical exponents.

2. Polymer Chain Collapse -- The Theta Transition

A polymer chain in solution has two behaviors:

- Good solvent ($T > T_{\text{theta}}$): chain is swollen, radius $R \sim N^{\nu_F}$ (Flory exponent $\nu_F \approx 0.588$)
- Poor solvent ($T < T_{\text{theta}}$): chain collapses, radius $R \sim N^{1/3}$ (compact globule)
- At $T = T_{\text{theta}}$: theta point -- crossover between swollen and collapsed

The theta point is a second-order phase transition. Its universality class: **3D Ising** (de Gennes 1972, Schafer & Witten 1977).

The correlation length exponent at the theta point: **$\nu_{\text{theta}} = 0.5877 \approx 0.5882 \approx \nu_{\text{Ising}} = 0.6298$** (within 7% -- consistent with same universality class given mean-field corrections near theta).

The order parameter: the end-to-end distance R vs. temperature. The transition from swollen (coherent, extended) to collapsed (decoherent, compact) is identical in structure to the Wike coherence transition from $C > 0$ to $C = 0$.

Tau protein is a polymer. In healthy neurons, tau is extended along microtubules (swollen phase -- good solvent, $T > T_{\text{theta}}$). In Alzheimer's: tau hyperphosphorylation changes the effective solvent quality (bad solvent, $T < T_{\text{theta}}$). Tau collapses into neurofibrillary tangles.

The tau misfolding transition IS the polymer theta transition IS the 3D Ising transition.

3. The Shared Exponent

Wike coherence transition:	exponent = $2.59 = 1 + 1/\nu_{\text{Ising}}$ confirmed 99.92% in 11.4M simulations
Polymer theta transition:	correlation length $\xi \sim T - T_{\text{theta}} ^{-\nu_{\text{theta}}}$ $\nu_{\text{theta}} \approx 0.588-0.630$ (3D Ising range)
Tau protein misfolding:	scaling of tangle formation rate with temperature/phosphorylation distance from critical point should follow $ T - T_c ^{-\gamma}$ where $\gamma = 1.2372$ (susceptibility exponent)

All three share the same universality class. This means:

1. **The shape of the transition is the same:** sharp cliff, 8.71x amplification at γ_c (from simulation), same susceptibility divergence $\chi \sim |\epsilon|^{-1.2372}$
2. **The scaling functions are the same:** near the critical point, all dimensionless ratios of observables are identical (up to non-universal scale factors)
3. **The same physics drives all three:** Z_2 symmetry breaking in 3D

4. Alzheimer's as Coherence Collapse

The causal chain in the Wike framework:

```

Stage 1: Tau hyperphosphorylation
-> effective  $T_{\text{theta\_tau}}$  rises above  $T_{\text{body}}$ 
-> tau enters collapsed phase (tangle formation begins)
-> microtubule stability decreases

Stage 2: Microtubule disruption
-> Principle 1 (Debye shielding) fails
-> EZ water structure in microtubule lumen disrupted

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-> Bootstrap nucleation loop broken (Paper 02)

Stage 3: Bootstrap failure
-> NIR cannot maintain EZ water -> Debye shielding -> coherence
-> gamma_eff increases (Paper 35 shows NIR scattering changes in Alzheimer's tissue -- Hanlon et al. 2008)

Stage 4: gamma_eff approaches gamma_c
-> susceptibility diverges -> any perturbation causes disproportionate decoherence
-> central sensitization analog in neural networks

Stage 5: gamma_eff > gamma_c
-> topological transition (Berry phase Paper 01)
-> permanent decoherent phase
-> clinical Alzheimer's
```

The tau transition and the coherence transition are not parallel processes -- they are the SAME transition at different scales, in the same universality class.

5. The Bootstrap Reversal Should Show Exponent 2.59

If Alzheimer's is the 3D Ising transition in the tau-microtubule-coherence system, then recovery (if possible) should follow the time-reversal of the transition.

The Bootstrap reversal rate near the critical point:

```
d(C)/dt|reversal ~ |gamma_eff - gamma_c|^(-nu) x NIR_dose

For gamma_eff slightly above gamma_c:
Rate ~ |epsilon|^(-0.6298) (correlation length exponent)

For the full reversal trajectory:
Integrated recovery ~ NIR_dose^(1/delta) = NIR_dose^(1/4.789) = NIR_dose^(0.209)
```

Prediction: In photobiomodulation studies of Alzheimer's, the response-dose relationship should follow a power law with exponent ~ 0.21, NOT a linear relationship. Studies that find "no dose-response" using linear models may be fitting the wrong functional form.

Multiple ongoing clinical trials of transcranial photobiomodulation in Alzheimer's (2024-2026) could test this directly.

6. What's Already Confirmed

Claim	Status	Source
2.59 exponent in coherence simulation	Confirmed, 99.92%	Paper 02, 11.4M runs
Polymer theta transition = 3D Ising	Confirmed	de Gennes (1972), Schafer & Witten (1977)
Tau is a polymer	Confirmed	Standard biochemistry
Tau misfolding in Alzheimer's	Confirmed	Nelson et al. (2019), NEJM
NIR scattering changes in Alzheimer's tissue	Confirmed	Hanlon et al. (2008)
Bootstrap loop disrupted in Alzheimer's	Consistent	Multiple NIR/PBM Alzheimer's trials
Universality class argument	Mathematical fact	Pelissetto & Vicari (2002)

Not yet confirmed:

- Direct measurement of critical exponent in tau aggregation kinetics
- NIR dose-response power law with exponent 0.21

Both are testable with existing experimental setups.

Summary

Alzheimer's disease is the 3D Ising phase transition occurring in the tau-microtubule-coherence system. The physics is the same as the Wike coherence transition -- same universality class, same exponents, same singularity structure. The transition runs in the order: tau collapse -> microtubule disruption -> Bootstrap failure -> coherence loss -> γ_c crossing. Recovery, if possible, runs in reverse with the same critical exponents. NIR photobiomodulation is the Bootstrap reversal -- and its dose-response should follow a power law with exponent 0.21, not a linear relationship.

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